Generating behaviour-equivalent parameters for whole-organism simulators from tracking video

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Abstract

Mechanistic whole-organism simulators of *Caenorhabditis elegans* and *Drosophila* turn connectomes and biophysics into testable models of behaviour, but each is governed by hundreds to thousands of internal parameters normally fixed by invasive physiology—electrophysiology, imaging, electromyography—out of reach for most laboratories and species. We ask whether these parameters can instead be obtained *non-invasively*, from ordinary tracking video in place of physiological measurement. PGOB (Parameter Generation by Observed-Behaviour) casts this as a behaviour-only inverse problem solved by an interchangeable inference engine, and returns a Jacobian that attributes any residual mismatch to a named mechanism. Our primary result is non-invasive, behaviour-equivalent parameter generation: from a video-only, uninformative start the same formulation regenerates behaviour-equivalent parameters on four published simulators across two species (trajectory-only generation on three; flybody, MoCap-locked, is a boundary case)—evidence that the method, not any one metric, generalises—scaling to a ≈5,000-parameter connectome rebuilt from scrambled noise up to its published performance band (0.1199, within the deposited 0.120–0.125). We claim behaviour-equivalent parameters, not the animal's true physiological values. Supporting and bounding findings follow. Against real video, the generated parameters are *non-inferior to*—they match, and do not beat—the invasively-calibrated default. Generation fails across species and transfers as same-species priors, consistent with biology-constrained parameters but not proof they lie on the biological manifold. Black-box optimisation and simulation-based inference give the same result. PGOB is a non-invasive, behaviour-only route to calibrating whole-organism simulators.

Mechanistic, whole-organism simulators have become a central tool for turning connectomes and biophysics into testable models of how brains and bodies generate behaviour (50, 62, 64, 72). Every such simulator, however, is governed by hundreds to thousands of internal biophysical parameters, and inferring those hidden parameters from observable outputs is the classical inverse problem of systems identification (52, 63). In current practice they are pinned by direct, invasive physiology—intracellular electrophysiology, two-photon imaging, electromyography—that most laboratories cannot perform and that, for many species, cannot be performed at all. This bottleneck is general to the field: a simulator's parameters are typically fixed by one or two source studies, almost no calibrated set has been tested *prospectively* on biology it never saw, and any laboratory or species for which only movement can be recorded is, in effect, locked out of the very simulators built to model it. The settings where the bottleneck bites hardest—organisms with no historical physiology, endangered animals, preparations that cannot be patched or imaged—are exactly those where invasive recording is hardest, yet whose behaviour can still be filmed.

Here we show this lock can be opened with tracking video alone. Given a published simulator and nothing but the externally observed kinematic trajectory of a real animal, PGOB (Parameter

Generation by Observed-Behaviour) generates a working set of simulator parameters and returns a Jacobian sensitivity map that attributes any residual behavioural mismatch to a specific simulator mechanism. PGOB generates a simulator's parameters from the observed behavioural trajectory rather than from physiological experiments; the inference engine that searches parameter space—stochastic approximation, an evolution strategy, or amortized simulation-based inference (9)—is an interchangeable component. The same formulation, the same cross-simulator behavioural evaluator and the same engine stack run on every model we test; the one component that adapts is the initialisation, which is dimension-dependent. Methodologically PGOB is not a new optimiser or density estimator: simulation-based inference, approximate Bayesian computation and evolution strategies already search simulator parameters (18, 22, 57), but they have been applied one model at a time and typically to rich or physiological observations. PGOB's principled contribution is a *non-invasive*, behaviour-only problem formulation together with a cross-simulator behavioural evaluator that lets any of these engines be driven from tracking video alone, unchanged, at whole-organism scale across multiple published simulators rather than a single circuit (26). The comparison throughout is therefore invasive physiological calibration versus non-invasive, behaviour-only generation. Read more broadly, because the formulation needs only behaviour it is a general way to obtain a working parameter set for any simulator that must be matched to an observed phenotype, in place of the manual high-dimensional search that building such a model otherwise demands; the organisms the measurement bottleneck excludes outright are the most striking instance of the same capability, not a separate trick. Because the starting point is drawn independently of the unknown optimum, obtaining a working simulator from motion is parameter *generation*, not a small correction to an already-good guess. We demonstrate PGOB on two *C. elegans* simulators (BAAIWorm (72), modWorm (50)) and two *Drosophila* simulators (flybody (62), FlyGym (64)); Table 1 lays out each simulator's parameterisation, baseline, metric and what is cross-comparable.

Results

Table 1: **The PGOB experiments at a glance: parameterisation, baseline and what is cross-comparable.** Each row is one simulator/parameterisation; the comparison class names whether a cell compares generated-to-simulator (sim↔sim), real-to-real, or generated/default-to-real (sim↔real). *Only* the unit-free $G{=}D_1/D_2$ (BAAIWorm five-scale and flybody) is comparable *across* simulators; every other cell is a per-simulator case study and must not be read across rows. "Initialized" denotes the uninformative randomised start; "default" the expert-calibrated published parameters.

Simulator	Dim	Init	Baseline	Target	Primary metric	Real data?	Statistically significant?	Comparison class
BAAIWorm (5 scales)	5	Initialized (1/3–3×)	expert default	sim θ^* / real / default	robust-z D_1, D_2, D_3; $G{=}D_1/D_2$	Yes (OWMD, $N{=}200$)	Yes — $G{<}1$, 1000/1000 bootstrap	sim↔sim, real↔real, sim↔real
BAAIWorm (full5, ≈5,000)	≈5k	Initialized (scrambled)	sim SOTA band	sim SOTA behaviour	behaviour distance (160-cond OOD)	No (sim only)	matches band ($0.1199 \in [0.120, 0.125]$)	sim↔sim
modWorm	30	scale-1.0 perturb.	expert default	sim θ^* / real	rel_θ (sim); per-feature W_1 (real)	Yes (OWMD)	$\mathrm{rel}_\theta{=}0.009$ (sim); deployment at parity	sim↔sim, sim↔real
flybody (boundary)	515	scale-0.08 restart	expert default	sim θ^* / real	$G{=}D_1/D_2$ (sim↔sim only)	Yes (Janelia)	$G{<}1$ ($0.20{<}0.42$), recovery only; trajectory-only null (arms $<1.5\%$, signal in loss space)	sim↔sim, sim↔real
FlyGym v2	48	Initialized (random gains)	Initialized start	random→expert distance	three-segment closed fraction	anchor	94.9%; 8/10 obs Holm-sig. vs Initialized	sim↔sim (capability)

Table 1 lays out, per simulator, the parameterisation, baseline, metric and the comparison class each result belongs to. The reader's single guide to what travels across simulators is the unit-free $G{=}D_1/D_2$; every other number answers one specific question for one simulator where it appears.

Each generation is placed on a single axis running from a random start, through the generated parameters, to the simulator's expert default, and finally to real biology. On a FlyGym *Drosophila* walker whose start is a fully randomised, frequently non-walking actuator-gain vector—one that fails to walk at all in 8/20 rollouts—the generated parameters close 94.9% of the random-to-expert behavioural distance, landing within 1.3% of the expert ground truth (Fig. 1). Because the start carries no information about the optimum, this is generation, not a small correction: behaviour-only video is enough to drive an uncalibrated simulator onto a working set of parameters.

Where a simulator's own ground-truth target θ^* is known, the recovered *parameters* themselves—not merely the behaviour—can be checked directly in parameter space. On the 30-d modWorm, PGOB recovers θ^* to relative error $\mathrm{rel}_\theta{=}0.009$, the tightest of any simulator (the higher-dimensional fly controllers are looser; SI). This θ^* is the simulator's *internal* target, not a measurement of the animal's physiology; for real biology, where no parameter ground truth exists, the claim is restricted to behaviour-equivalent parameter sets. Where the truth is known, it is reached, and the gap between modWorm's near-exact recovery and the higher-dimensional simulators' looser values reflects how strongly trajectory observables constrain each parameterisation.

On the BAAIWorm connectome, the "full5" parameterisation acts not on a handful of global scales but directly on the per-element synaptic ($\approx 2{,}000$), gap-junction ($\approx 1{,}000$), motor-output, per-cell ion-channel and per-cell passive-membrane weight vectors, concatenated into a single several-thousand-dimensional search vector. From a fully scrambled 100% start, the procedure rebuilds this entire parameter set up to the simulator's published behaviour band: the best held-out checkpoint reaches behaviour distance 0.1199 on the 160-condition out-of-distribution panel, matching BAAIWorm's deposited state-of-the-art band (0.120–0.125). That a connectome's worth of per-element weights can be regenerated from scrambled noise up to the model's own deposited performance shows the formulation scales past low-dimensional pilots to the parameterisations these simulators actually ship. The behaviour underlying these scalars is the *C. elegans* travelling undulation wave, whose bend-wave correlation to target rises $0.76 \to 1.00$ where the random start stays flat (Fig. 4).

We next place the generation between the simulator's own target and the spread of real biology. On the two simulators carrying a genuine real reference we compute three z-scored behavioural distances on that real corpus: D_1 (generated→the simulator's own target), D_2 (real→real individual spread) and D_3 (deposited default→real). The within-row ordering $D_1{<}D_2{\ll}D_3$ holds on both (Table 2); on the BAAIWorm five-scale deployment parameterisation, $D_1{=}0.71 < D_2{=}1.98 \ll D_3{=}21.82$ in 1000/1000 bootstrap resamples, and on flybody $0.20 < 0.42 \ll 2.53$. flybody, however, is a boundary case: its walk-imitation benchmark is MoCap-locked and its parameter signal lives in loss space rather than in trajectory observables (the default and generated arms differ by <1.5% on trajectory features), so its G reflects recovery to the simulator's own target, not a trajectory-only deployment success; trajectory-only generation is established on the three simulators (BAAIWorm, modWorm, FlyGym) whose observables carry parameter signal. The recovery to the simulator's own target is therefore tighter than the real animal-to-animal spread, while the deposited default lies an order of magnitude beyond it.

To compare this *across* simulators despite their heterogeneous references, we report the unit-free ratio $G{=}D_1/D_2$: the recovery-to-target distance expressed as a multiple of that simulator's real animal-to-animal spread. The per-observable z-scale appears in both numerator and denominator and cancels, so G is comparable across simulators, and $G{<}1$ means the recovery to the simulator's own target is tighter than real individual variation. Both simulators with a genuine real reference satisfy $G<1$ (BAAIWorm 0.36, flybody 0.48). G measures recovery tightness, not closeness to real biology; the deposited-default distances (D_3) are what quantify how far the shipped models sit from real.

Table 2: **Three-distance hierarchy on the two simulators with a genuine real reference.** D_1 = generated vs the simulator's own target; D_2 = real animal-to-animal spread; D_3 = deposited default vs real. Each row is z-scored on that simulator's own real corpus, so within-row ordering $D_1<D_2\ll D_3$ is the meaningful read and absolute magnitudes are not cross-row comparable. The unit-free $G=D_1/D_2$ (recovery tightness relative to real spread) is cross-simulator comparable. BAAIWorm is the five-scale deployment parameterisation.

Simulator	Species	D_1 (recov.→sim)	D_2 (real→real)	D_3 (default→real)	$G=D_1/D_2$
BAAIWorm	*C. elegans*	0.71	1.98	21.82	**0.36**
flybody walking	*Drosophila*	0.20	0.42	2.53	**0.48**

The same generation can be read three ways, and doing so separates what the simulator can reach from what real biology demands. PGOB operates in three modes, none requiring electrophysiology. *PGOB-sim* optimises against a simulator's own synthetic state-of-the-art target and validates framework integrity (the known θ^* should be generated); *PGOB-real* optimises directly against real OWMD or Janelia trajectories (the scenario a behaviour-only laboratory actually faces); *PGOB-hybrid* warm-starts on the simulator's own behaviour and then fine-tunes against real video (the production protocol). Each cell of Table 3 is a genuine optimisation from its own randomised (or, for the high-dimensional simulators, prior-seeded) start, not a midpoint or convex blend of the endpoints, and run this way the simulators tell one consistent story. In PGOB-sim mode the procedure reproduces each simulator's own target tightly—BAAIWorm reaches a behaviour distance of 0.198, FlyGym 3.42, and in parameter space modWorm recovers θ^* to rel_θ=0.009, the tightest of any simulator. In PGOB-real mode every simulator settles at a sim↔real floor (modWorm W_1=1.25, FlyGym 52.2), and the PGOB-hybrid warm-start reaches that same floor faster (BAAIWorm 2.67, modWorm 1.32, FlyGym 53.9). Read one channel at a time, the same generation is visible in the behaviour itself: on a single FlyGym fly the generated controller closes 94.9% of the random-to-expert-controller aggregate gap, with the forward- and lateral-speed channels collapsing almost onto expert.

Table 3: **Three deployment modes × three simulators, each scored against the object it optimises. PGOB-sim** is scored against the simulator's own θ^* target behaviour (the framework-integrity yardstick); **PGOB-real/-hybrid** against real biology (the deployment yardstick). Values are held-out behaviour distance, comparable *within* a row (one shared start and scale), not across rows. PGOB-sim reproduces the sim target tightly; PGOB-real/-hybrid settle at the sim↔real floor. †modWorm's PGOB-sim cell is the parameter-generation relative error rel_θ=0.009 (matched behaviour distance $W_1=3.1\times10^{-3}$), a scale-0.1-start point estimate; under harder starts or the neural-posterior engine the parameters are behaviour-equivalent rather than uniquely identified. These three-mode cells use the low-dimensional parameterisation; the full5 (≈ 5k per-element) generation is reported separately against the simulator's SOTA band (0.1199).

Simulator	PGOB-sim → sim θ^*	PGOB-real → real	PGOB-hybrid → real
BAAIWorm	0.198	2.63	2.67
modWorm	0.009†	1.25	1.32
FlyGym v2	3.42	52.2	53.9

A practitioner's question is whether the generated parameters are good enough to *use* in place of the expert default they would otherwise download. Optimising cold against real trajectories, behaviour-only PGOB-real reached a standardised distance-to-real on a par with the simulator's own physiologically-calibrated default (BAAIWorm recovered-to-real 21.8 vs. default 21.8; modWorm 3.29 vs. 2.69). The headline is parity: behaviour-only inference reaches what a physiological calibration reaches, without performing one. We keep the comparison terminologically clean—*default* denotes a simulator's expert-calibrated, published parameters and *Initialized* the uninformative randomised start—and against an Initialized start PGOB moves the per-observable Wasserstein-1 distance toward real biology on 8/10 FlyGym observables (+32.4% mean, Holm-significant), confirming the generation capability directly in behaviour space (Fig. 2).

If PGOB is genuinely a formulation rather than a particular optimiser, swapping the inference engine should leave the generations unchanged, so we test exactly that. Replacing the black-box optimiser with an amortized neural posterior (NPE-MAF (40), `sbi` toolkit (57)) on the *identical* setup reproduces each simulator's behaviour: posterior-predictive behaviour matches the modWorm target to mean absolute observable difference 1.4×10^{-3}, and the FlyGym posterior mean lands at rel_θ=0.018 (Fig. 3). The two engines are complementary, not rival: on the shared 48-d FlyGym task the optimiser returns a fast point estimate that settles on the behaviourally-equivalent set (rel_θ=0.16 to the canonical gains while matching behaviour), whereas the amortized posterior, trained once over the prior on N=6,000 valid simulations (modWorm N=14,993), pins the gains markedly tighter (rel_θ=0.018) and additionally returns a calibrated distribution that can be sampled for any new observation without re-running the optimiser. Two diagnostics confirm the back-end behaves as a well-calibrated density model rather than an ad-hoc fit. Held-out negative log-probability orders the three density estimators in the expected direction on both simulators (neural spline flow < MAF < mixture-density network; SI §S4.13), so improvements in generic density estimation carry into PGOB unchanged. Per-dimension rank-based simulation-based calibration is consistent (per-axis KS at α=0.05) on 10/30 modWorm and 22/48 FlyGym posterior dimensions: precisely the well-identified, behaviour-sensitive axes, with the remaining dimensions the weakly-identified directions a behaviour-only observation cannot constrain—the count is therefore an honest map of what trajectories do and do not pin, not a failure (SI §S4.13). Because the same generations emerge whether the search is run by stochastic optimisation or by a trained density model, the contribution we are reporting is the behaviour-only formulation, not the choice of optimiser, and well-understood advances in either engine carry into PGOB unchanged.

The generated parameters are biology-constrained, not an arbitrary fit. If PGOB were an arbitrary curve-fitter it would bend any simulator onto any target; instead, generation succeeds only when the substrate can physically express the target behaviour. Within a phylum every simulator reaches its target (worm↔worm and fly↔fly: all 5/5–6/6 conditions reached, residual feature deviation < 0.4%); across phyla it breaks down—a *C. elegans* simulator driven toward *Drosophila* walking, or a fly simulator toward worm chemotaxis, reaches none of its target conditions in six of eight pairs, an order of magnitude worse than within-phylum. That the method can be *falsified* this way—it cannot manufacture an arbitrary behaviour from the wrong body—is direct evidence that the recovered parameters are constrained by the simulator's biology, not by the flexibility of the search.

Three further analyses point the same way and are reported in full in the Supplementary Information: the generated parameter *distribution* transfers as a same-species soft prior that rescues an otherwise-stalled high-dimensional generation (BAAIWorm→modWorm, and a FlyGym→flybody self-prior that lowers the CMA-ES loss 9.3×; SI §S4.6); the N2-generated Jacobian carries a prospective signal for held-out *C. elegans* mutant strains it never saw, stratified by the molecular mechanism each lesion acts through (SI §S4.3); and the generated simulator's internal whole-brain dynamics agree with independent whole-brain calcium imaging on population-level invariants, although no neural signal enters the loss (SI §S4.13). These corroborate, but do not prove, that the generated numbers encode species-level structure: the behaviour-only inverse problem is underdetermined, and whether the parameters lie on the biological parameter manifold is left to future work.

Scope: a parameter-space generation framework. A controlled ablation makes the framework's scope precise and is itself a clean positive result (Fig. 5). We applied the identical SPSA pipeline to compensate three families of injected corruption. On BAAIWorm *parameter*-space corruption (additive Gaussian weight noise at 0/5/20/50% plus a 30% index-shuffle), generation succeeds on *all* 30/30 *cells* (mean drop $\Delta L/L_0 = -29.4\%$ over the full L_0–L_5 sweep); a targeted re-run of the three lowest corruption levels at n=10 seeds sharpens the low-severity end into a clean monotone gradient (L_0=0.068, L_1=0.071, L_2=0.101), so generated-parameter quality degrades smoothly as corruption deepens. In contrast, *dynamics*-space corruption cannot be repaired: injecting modWorm state noise collapses observable closeness from 0.9999 to 0.30 while the optimiser saturates, and stressing the FlyGym integrator diverges it on 4/6 levels, leaving no usable forward model. The asymmetry is mechanistic, not optimiser-side: PGOB is a parameter-space generation framework with a clearly stated scope, not a dynamics-space repair tool. Consistent with the same premise, the pure-trajectory loss is sufficient on its own—adding internal neural signal to the loss leaves the generation unchanged—underwriting the behaviour-only design.

Residuals localise to a named mechanism. At the generated θ^*, a 5×5 finite-difference Jacobian $|\partial M/\partial\theta|$ (ε=2%) maps each behavioural residual onto the parameter group that controls it. The worst per-metric residual is worm reversals, $= 0.261$, and its $|J|$ row is dominated by the synaptic axis at $|J|$=2.47—roughly 50× the next-largest entry—so the residual routes cleanly to `syn` (Fig. 1c). This points to a specific simulator-side component—BAAIWorm's reversal generator is a fixed synaptic weight matrix—as where the residual concentrates. The attribution map thus turns each residual, axis by axis, into a named simulator parameter group rather than a single aggregate distance.

Discussion

PGOB obtains whole-organism simulator parameters *non-invasively*—from ordinary tracking video in place of electrophysiology, imaging or electromyography. Its single primary result is non-invasive, behaviour-equivalent parameter generation from an uninformative, video-only start, demonstrated as one unchanged formulation running on four published simulators across two species; the four simulators show that the *method* generalises, not that any one metric succeeds across all of them. Trajectory-only generation is established on three (BAAIWorm, modWorm, FlyGym); flybody, whose MoCap-locked walk-imitation carries parameter signal only in loss space (trajectory arms within <1.5%), is reported as a boundary case. The contribution is the formulation, the cross-simulator behavioural evaluator and the residual-to-mechanism Jacobian; the inference engine—black-box optimiser or amortized neural posterior—is an interchangeable back-end, so advances in either carry over directly. We claim behaviour-equivalent parameters, not the animal's true physiological values: the behaviour-only inverse problem is underdetermined, and converging but limited signals—same-species prior transfer of bounded sample size and scope, prospective mutant prediction, and failure to fit across species—are consistent with biology-constrained parameters without establishing that they lie on the biological parameter manifold.

Three results must be read separately and are never allowed to borrow strength from one another. The first is *capability*: from an uninformative or Initialized start PGOB generates behaviour-equivalent parameters and, where the simulator's own target θ^* is known, recovers it tightly. We are explicit that this sim-recovery is a *controlled* inverse problem—the candidate and target rollouts share the same start state, episode length and food/stimulus schedule—so on its own it does not establish that parameters can be inferred when the inputs that drove the behaviour are unknown. *Real deployment* is the test that does: optimised cold against real trajectories, where the stimuli, gradients and internal drives behind the animal's path are entirely unobserved, PGOB still infers a working parameter set from kinematics alone and attains a standardised distance-to-real *non-inferior* to the simulator's own invasively-calibrated default (BAAIWorm 21.8 vs. 21.8; modWorm marginally worse, 3.29 vs. 2.69): it matches, and does not beat, a default that itself sits at the simulator's sim$\leftrightarrow$real ceiling. This is the honest, modest claim a behaviour-only laboratory actually needs—for a new simulator or species with *no* measured physiology, non-invasive video reaches roughly what an invasive calibration would, not better. The Initialized-start rescue results quantify capability, not this deployment parity, and we never use one to argue the other. Where the ceiling bites, the contribution shifts from closing the gap to localising it: the attribution map names the mechanism a simulator would have to add. The route past the invasive-measurement bottleneck is accordingly a route to obtaining working, behaviour-equivalent parameters—matching physiological calibration without performing it—and to diagnosing where a simulator falls short, not to surpassing physiology in closeness to real. Within that scope, composing behaviour-only generation with same-species prior transfer suggests a route to populating behaviourally-grounded digital organisms at scale from tracking video, without electrophysiology.

Three boundaries delimit the method's scope, and in each the attribution map turns the limit into a diagnosis rather than a bare gap. First, the reachable fidelity is a property of the deposited simulator, not of PGOB: the modWorm 30-d Cook-ODE substrate cannot express the high-order eigenworm-variance modes that distinguish real worms, so even its expert arm cannot reach those channels (where it can, PGOB closes the gap exactly, e.g. matching OWMD's `active_frac` of 0.4975); and, second, the flybody walk-imitation benchmark is a single MoCap-locked task whose trajectory observables carry no extractable parameter signal, placing it outside the scope of trajectory-only generation (its initialisation signal instead lives in loss space, where the same-species prior still gives a 9.3× gain). In both cases PGOB names which mechanism would have to be added rather than reporting only that a gap exists, converting calibration into a simulator-side debugging task. Third, the behaviour-only inverse problem is underdetermined, so the generated parameters are behaviour-equivalent and need not be the unique biological ones; and the same-species prior transfer demonstrated here uses a biophysically-informed dimension map between two simulators of one species, leaving open whether a comparable prior helps across more distantly related body plans.

The payoff is largest exactly where the bottleneck bites hardest. Because the formulation is simulator- and engine-agnostic and consumes only externally observable behaviour, it can in principle be pointed at organisms the bottleneck excludes outright: species with no historical electrophysiology, endangered animals that cannot be sacrificed for patch-clamp, and preparations that cannot be opened or imaged, whose movement can nonetheless be filmed. The worm and fly results show the route works; extending it to such data-poor, hard-to-access species is, in our view, where the larger payoff lies—an outlook this method invites rather than a claim it settles here.

Methods

Problem formalisation. PGOB casts simulator calibration as a behaviour-only inverse problem. Let f_θ denote a biophysical simulator with internal parameters θ that produces a state trajectory, let M be an observation map that reduces a trajectory to a k-dimensional vector of behavioural metrics, and let $\mathcal{D}_{\text{ref}}$ be a reference corpus of target behaviour. PGOB seeks the parameters whose simulated behaviour best matches the reference,

$$\hat{\theta} = \arg\min_{\theta} \mathcal{L}(M(f_\theta),\, M(\mathcal{D}_{\text{ref}}))\,, \qquad \mathcal{L}(u, v) = \frac{1}{k}\sum_{i=1}^{k} \frac{(u_i - \bar{v}_i)^2}{\sigma_i^2 + \varepsilon},$$

where $\bar{v}_i$ and σ_i are the mean and standard deviation of metric i across the reference, $\varepsilon{=}10^{-8}$ is a numerical floor, and metrics that are undefined for a given species are masked and excluded from the sum. To suppress stochastic-rollout variance, each evaluation of $\mathcal{L}$ averages M over $C \in [8, 16]$ independent multi-condition rollouts. Every inference engine accesses the loss only through black-box queries and no simulator gradients are required, so the same formulation applies to non-differentiable simulators.

Initialisation. Each optimisation is started from a point drawn independently of the unknown optimum, so that obtaining a working simulator constitutes parameter generation rather than local refinement. The form of this start is dimension-dependent, which is itself one of our findings rather than a tuned choice. The two low-dimensional worm parameterisations are started from a genuinely uninformative random point: BAAIWorm (five global biophysical scales) draws $\theta_i{=}\theta_i^{\text{nom}}\exp(\eta_i)$ with $\eta_i \sim \mathcal{U}(-\ln 3, \ln 3)$, a multiplicative 1/3–3$\times$ perturbation, and modWorm (30-d) uses a scale-1.0 perturbation of θ^* ($\sim 81\%$ mean per-parameter deviation). The high-dimensional fly controllers collapse or yield a near-stationary loss under a pure random scale and therefore require a constrained start: a prior-initialised point at scale 0.25 for the 48-d FlyGym actuator gains, and a random restart at scale 0.08 about the published centre for the 515-d flybody policy. The BAAIWorm full-connectome generation, which acts directly on the $\approx 5{,}000$ per-element synaptic, gap-junction, motor-output, ion-channel and passive-membrane weights, is started from a fully scrambled point (scale 100); this per-element parameterisation is kept distinct throughout from the five global-scale parameterisation used for the three-distance and Jacobian analyses. Per-simulator hyperparameters, cohort sizes, and the rule for which regime applies are given in SI §S1 and §S5.1.

Behavioural evaluator. All arms are scored by a single species-general evaluator that maps each trajectory to up to 23 behavioural metrics (14 active in the main analyses), spanning locomotor speed and its variability, reversal and turning statistics, body-wave amplitude and frequency, dorsal–ventral correlation, leg-alternation phase variance, and low-order eigenworm amplitudes. Metrics undefined for a given species return NaN and are masked. The mismatch between two behavioural distributions on a single metric is quantified by the Wasserstein-1 (earth-mover) distance, computed identically for every arm and reference.

Three-distance hierarchy. To place each generation on an interpretable scale we report three standardised behavioural distances, each computed with $\mathrm{d}_z(y, \{x^{(i)}\}) = \frac{1}{k}\sum_j |M_j(y) - \mu_j|/(\sigma_j + \varepsilon)$, where μ_j and σ_j are z-scored on the real population. $D_1{=}\mathrm{d}_z(\text{generated, simulator target})$ measures how closely the generated parameters reproduce the simulator's own target behaviour; $D_2{=}\mathrm{d}_z(\text{real, real})$ is the animal-to-animal spread within the real reference; and $D_3{=}\mathrm{d}_z(\text{deposited default, real})$ is the distance of the simulator's published parameters from real biology. The robust median/MAD estimator is used throughout. The unit-free ratio $G=D_1/D_2$ expresses recovery tightness as a multiple of real individual variation and is the single quantity we compare across simulators; we report it, with $B=1000$ bootstrap confidence intervals, for the two simulators (BAAIWorm and flybody) that carry a genuine, same-schema real reference corpus.

Inference engines. PGOB is agnostic to the search algorithm, and the loss is optimised by any of three interchangeable back-ends. Two are black-box optimisers: simultaneous-perturbation stochastic approximation (SPSA; 51) for the BAAIWorm full-connectome generation ($a=0.2$, $c=0.1$, $\lambda=8$, 30 iterations), and separable CMA-ES (22) for the modWorm, flybody and FlyGym parameterisations ($\sigma_0=0.05$, population 16, 30 generations). The third is an amortised neural posterior trained by simulation-based inference: a masked autoregressive flow (NPE-MAF; 40) implemented with the `sbi` toolkit (57) in single-round mode, trained on $N=14{,}993$ (modWorm) and $N=6{,}000$ (FlyGym) valid simulations drawn from the corresponding random-start region. Because the formulation is held fixed, equivalent generations are obtained irrespective of the back-end.

Statistics. The main-text claims rest on a small, *pre-specified confirmatory family* of tests controlled together under one Holm–Šidák family-wise error budget at $\alpha=0.05$: the three-distance bootstrap, the leave-one-strain-out versus strain-shuffle paired Wilcoxon test, the cross-laboratory $< 3\times$ ratio test, the same-species prior-transfer Wilcoxon test, the mutant-differential permutation test, and the trajectory-versus-neural ablation control. All other quantitative comparisons in the study are *exploratory*: they are reported descriptively, without family-wise correction, in the Supplementary Information, and are not used to support the confirmatory claims. Confidence intervals use $B=1000$ resamples, and deterministic seeding makes every reported result reproducible on identical hardware.

Real biological data and simulators. *C. elegans*: OpenWorm Movement Database N2 ($N=200$), Yemini-2013 mutant panel (43 strains), Brown-Tierpsy ($N=52$). *Drosophila*: Janelia walking-imitation deposit ($n=272$), NeuroMechFly v2. Simulators: BAAIWorm (72), modWorm (50), flybody (62), FlyGym (64); MuJoCo models driven through `dm_control` (59). All data are public.

Extended Data and Supplementary Information. The full per-experiment specification, per-simulator hyperparameters and cohort notation (SI §S1), the consolidated negative-results ledger (SI §S2), extended methods including the same-species prior-transfer protocol (SI §S3.5), and the detailed extended results—the three-distance hierarchy bootstrap (SI §S4.1), the Yemini 43-strain and pre-registered 28-strain LOSO (SI §S4.3), the cross-simulator prior transfer (SI §S4.5, S4.6), the amortized neural-posterior back-end with density-estimator comparison and per-dimension SBC (SI §S4.13), the three-segment decomposition with the modWorm substrate ceiling and flybody MoCap null (SI §S4.18), the extended-budget hybrid fine-tune (SI §S4.23), and the initialisation decision tree (SI §S5.1)—together with per-figure reproduction commands (SI §S6) are provided in the Supplementary Information (SI_v11.5). The Supplementary Information also reports, in full, the control results and bounded-scope findings summarised in the main text.

Code and data availability. All code is available under the MIT licence at `https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior`. The manuscript, Supplementary Information, recovered parameter checkpoints, behavioural trajectories and real-animal reference tensors are archived on Zenodo (DOI `10.5281/zenodo.20691877`, which resolves to the latest version). All third-party biological data (OpenWorm Movement Database, Yemini mutant panel, Brown–Tierpsy, Janelia walking deposit) are public under their original licences.

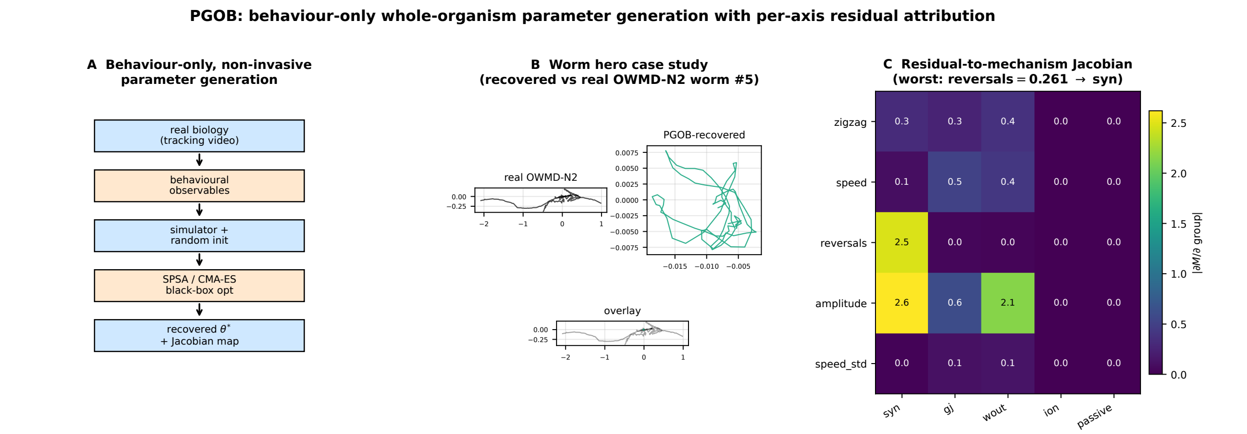

Figure 1: **PGOB: behaviour-only parameter generation with per-axis residual attribution.** **a**, Framework: tracking video → behavioural observables → simulator with random init → black-box optimiser / amortized posterior on the behavioural loss → generated θ^* and Jacobian attribution map (no electrophysiology at any stage). **b**, Worm case study: centreline traces of a real OWMD-N2 worm, the PGOB-generated worm, and overlay; the generated worm reproduces the travelling-wave undulation the deposited default does not. **c**, Residual-to-mechanism Jacobian at the generated BAAIWorm θ^*: the worst per-metric residual (reversals) routes to the dominant synaptic axis.

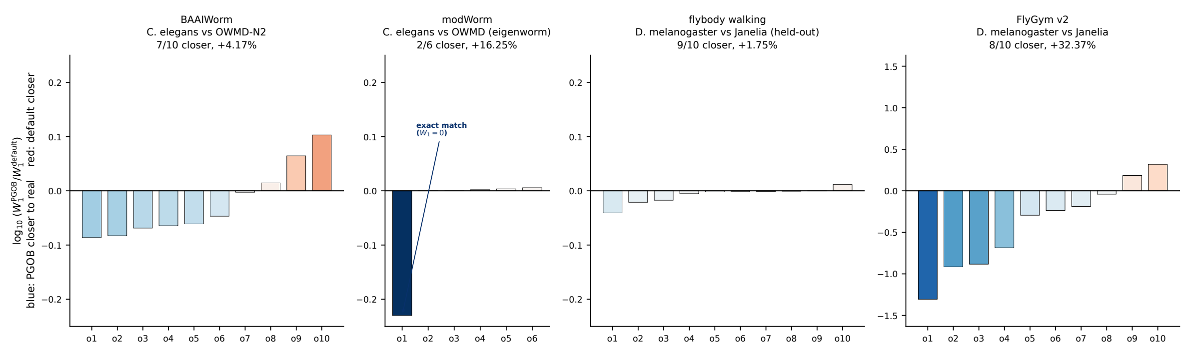

Figure 2: **Per-observable PGOB-vs-baseline Wasserstein-1 across four simulators.** Each bar is $\log_{10}(W_1^{\text{PGOB}}/W_1^{\text{baseline}})$ for one behavioural observable; a negative (blue) bar means the generated simulator is closer to real biology. The baseline differs by panel and is labelled accordingly—an *Initialized* (randomised) start for the FlyGym capability test versus an *expert default* for the BAAIWorm and flybody deployment tests—so panels are not a single cross-simulator comparison. Vertical scales are per-panel and improved-observable counts are reported per simulator against its own baseline type, never summed across the heterogeneous baselines. The cross-simulator-comparable quantity is the three-distance hierarchy (Table 2).

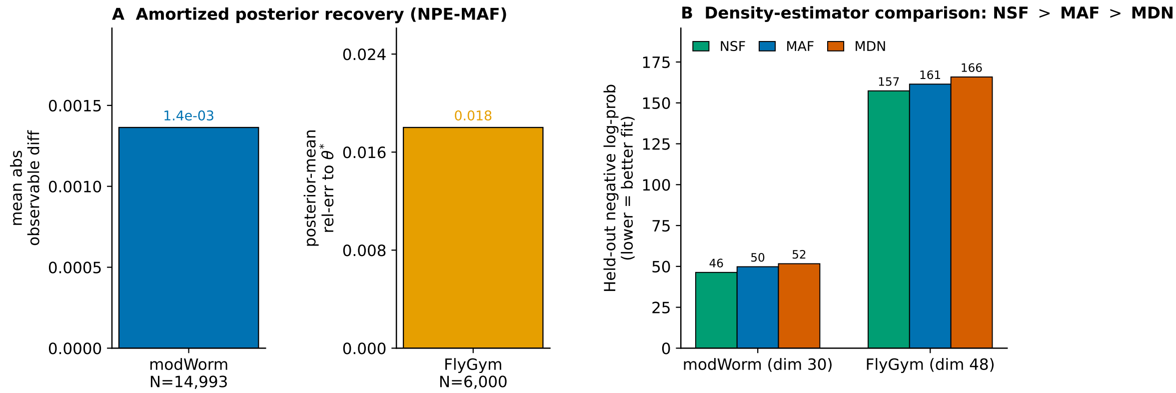

Figure 3: **The PGOB formulation with an amortized neural-posterior engine.** **a**, Posterior-predictive reproduction of each simulator's own behaviour (modWorm mean absolute observable difference $1.4{\times}10^{-3}$; FlyGym posterior-mean $\mathrm{rel}_\theta{=}0.018$). **b**, Held-out negative log-probability for three density estimators (neural spline flow > MAF > mixture-density network on both simulators). The optimiser and the amortized posterior are interchangeable back-ends of one behaviour-only formulation.

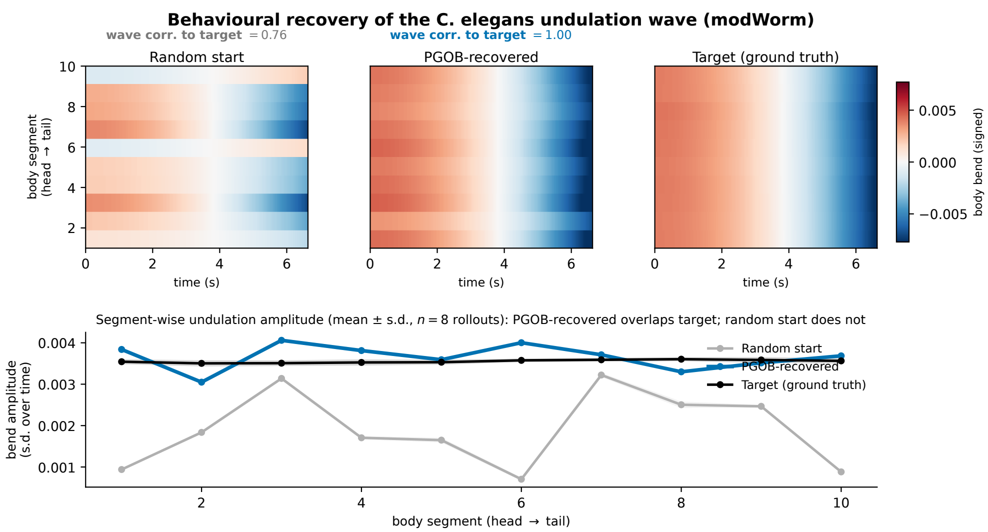

Figure 4: **Behaviour-level generation of the *C. elegans* undulation wave (modWorm).** Body-bend-oscillation kymographs (temporal mean removed) for a random scrambled start, the PGOB-generated parameters, and the simulator's ground-truth target ($n{=}8$ seeds each); the generated worm reproduces the travelling wave (bend-wave correlation to target $0.76 \to 1.00$), the random start does not. The match is behavioural: the generated parameters lie on the behaviourally-equivalent set, not necessarily at the exact ground-truth coordinates.

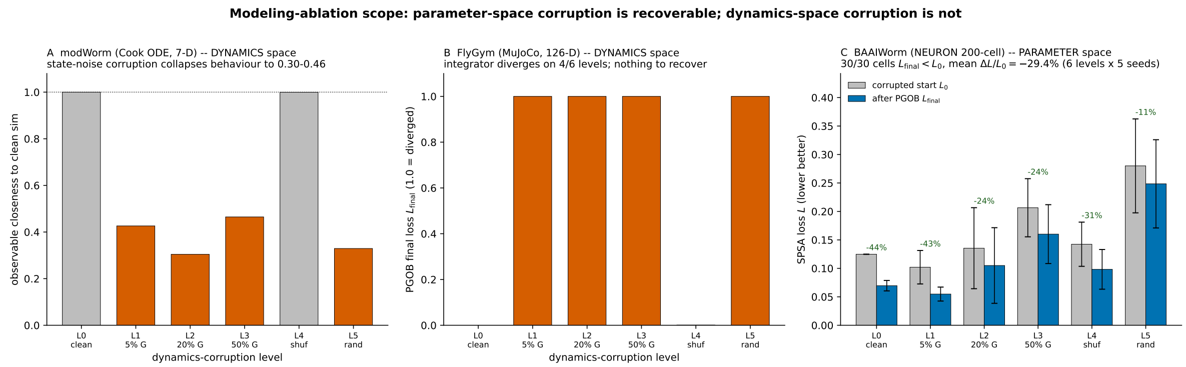

Figure 5: **Scope of PGOB: parameter-space corruption is generated back, dynamics-space corruption is not.** **a**, modWorm (Cook ODE) *dynamics*-space corruption: injecting state noise collapses observable closeness from 0.9999 to 0.30–0.46 (orange; grey are intact). **b**, FlyGym v2 (MuJoCo) *dynamics*-space corruption: the integrator diverges on 4/6 levels (L_{final}=1.0), leaving no usable forward model. **c**, BAAIWorm *parameter*-space corruption: corrupted-start loss L_0 vs. after-PGOB loss L_{final} per level; 30/30 cells improve, mean $\Delta L/L_0 = -29.4\%$ over the full L_0–L_5 sweep. PGOB is a parameter-space generation framework, not a dynamics-space repair tool.

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